Learning Module 53: Calorimetry

Started: Mar 19 at 9:45am

Quiz Instructions

This assignment contains 8 multiple choice questions. It allows for three attempts and the highest score is applied. This assignment is not timed and can be completed in multiple sittings. Read each question completely and thoroughly. Additional resources may be referenced. Please remember that OnRamps students are considered students at The University of Texas at Austin and are subject to the University’s academic integrity policies. Each student in the course is expected to abide by the University’s Honor Code.

Calorimetry.

Calorimetry is a wonderfully useful tool that is made possible because of the first law of thermodynamics. We use calorimetry to tell us how much heat energy is stored in a system. You are most familiar with it because calorimetry is the experimental technique that tells you how many calories are in your food. Every package of food has associated with it a caloric content, like 12 calories per serving.

This module presents both the theoretical foundation and the experimental design associated with the study of calorimetry.

[Video Transcript (https://onramps.instructure.com/courses/4073013/files/292196467?wrap=1)](https://onramps.instructure.com/courses/4073013/files/292196467?wrap=1) (https://onramps.instructure.com/courses/4073013/files/292196467/download?download_frd=1)

(Remember that E and U are used interchangeably in the video to describe internal energy)

In this video, Dr. Laude talks about two kinds of calorimetry, **bomb calorimetry** and **coffee cup calorimetry**. These phrases refer to the experimental device used to measure heat change and earn those names either because the device looks like a bomb or looks like a coffee cup.

Besides the interesting names, what is the difference between bomb and coffee cup calorimetry?

Bomb calorimeter equation. Recall that internal energy is defined as

$$\Delta U = \Delta E = q + w = mC\Delta T + P\Delta V$$

Now imagine designing an experiment in which the calorimetry took place in a closed system that could not change its volume.

This would mean that

$$w = P\Delta V = 0$$

and

$$\Delta U = \Delta E = q + 0 = mC\Delta T$$

State simply, in a closed system in which there is no volume change

$$\Delta U = \Delta E = q_v$$

(the subscript v means that the volume doesn't change.)

So if we want to know how much heat energy is in a cup of cereal or a pound of lettuce, we need to do the experiment in a device that can't change volume.

The device that does calorimetry without changing volume is called a **bomb calorimeter.**

Coffee cup calorimeter. In the real world, internal energy changes occur in an open system. The earth is an open system, your body is an open system. This means that the internal energy change is NOT equal to the heat energy.

Remember, though, that the definition of enthalpy change is:

$$\Delta H = \Delta U + P\Delta V$$

so in an open system where there is PV work done because the volume can change:

$$\Delta H = \Delta U + P\Delta V = q + w + P\Delta V = q - P\Delta V_{ext} + P\Delta V = q$$

That is a lot of canceling to leave us with a very simple equation:

$$\Delta H = q_p$$

(the subscript p means that the pressure doesn't change.)

Or simply, in an open system where the pressure doesn't change (the atmospheric pressure) but the volume does, the heat change in the system is a change in enthalpy, ΔH.

Without going into too much detail, ΔH has a correction factor for the fact that some of the heat energy in an open system is being used to do work against the external atmospheric pressure. You might not appreciate it right now, but the next time someone tells you to get up off the couch and get some work done, just let them know that you are already working hard against the atmospheric pressure pounding against your body.

Summary. Realistically I don't expect a full appreciation of the derivations show above for a bomb and a coffee cup calorimeter. But you should know these simple statements:

Bomb calorimetry is at constant volume so, to calculation heat change:

$$\Delta U = \Delta E = q_v = mC\Delta T$$

Coffee cup calorimetry is at constant pressure, so to calculation heat change:

$$\Delta H = q_p = mC\Delta T$$

Question 1 1 pts

You toss the leftover food from a camp out into the campfire and it burns. Which of the following changes in state functions describe the heat released?

- ΔU
- ΔS
- ΔG
- ΔH

Bomb Calorimetry

The point of calorimetry is to have a way of measuring the heat content in a system. But that is now so easy to do directly. Remember that exploding balloon in the video above? We wanted to know the **energy released as heat** by that balloon.

$\Delta H = mC\Delta T$

ΔH = heat change of system
m = mass of system
C = specific heat
ΔT = temperature change

But this is hard to measure directly because the heat is given off in every direction, so any ΔT measured is likely inaccurate. Moreover, if we don't happen to know the heat capacity of the hydrogen gas, then there is another part of the $mC\Delta T$ equation that is unknown.

How do we solve this problem? The answer is to take advantage of the first law of thermodynamics.

If we were to surround our system of interest with a second container for which we can more easily measure q, then the first law of thermodynamics says that:

$$q_{sys} = -q_{surr}$$

In other words, all the heat in that exploded hydrogen balloon has to be transferred to the surroundings, and if we set up the calorimetry experiment correctly, it is much easier to measure q_{surr} and, because of the first law, find q_{sys} indirectly.

Do this, we construct the bomb calorimeter shown below. Note its features:

1. A heavy-duty container houses the system so that when a combustion reaction occurs, there is no change in volume.
2. The surroundings need to have an easily measured ΔT and a known C. An obvious choice is water, which has a high C so that there isn't a huge change in ΔT, and its C is well known: $C_{H2O} = 4.18$ J/g°C.

Look at what we can do with our original calorimetry equation

$$mC\Delta T_{sys} = q_{sys} = -q_{surr} = -mC\Delta T_{H2O}$$

Note that we don't have to know anything about the system to find its heat content as long as we know the mass of the water and have a way of creating a uniform temperature in the water. (This is why a stir bar is used.)

Example. A potato chip explodes in a bomb calorimeter surrounded by 200 grams of water that increases in temperature from 25°C to 27°C. How much energy is in the potato chip?

$$q_{surr} = -mC\Delta T_{H2O} = 200 \text{ grams} \times 4.18 \text{ J/g°C} \times 2\text{°C} = -1600 \text{ J}$$

Therefore, the energy in the potato chip (q_{sys}) = 1600 J or 1.6 kJ

A few additional things to remember.

- This is a bomb calorimeter so you are making a constant volume measurement of ΔU:

$$\Delta U = \Delta E = q_v$$

- The signs associated with calorimetry can be confusing, but you shouldn't worry too much. When we perform a calorimetry measurement to find the heat content in a potato chip, we don't really care about "be the system". We just want to know if we eat too many, will we have to go run around the block to lose a little weight. So most answers you give will have a positive sign that doesn't care about the direction of energy flow.

- If you have ever held a cup of hot coffee in your hand, you know that when heat comes into the water, it will then also leave and go to the cup and then to your hands. In other words, the ΔT of the water doesn't account for all the energy that leaves the system. Some will go into the calorimeter itself. To account for this, the more accurate description of a calorimetry equation is:

$$mC_{sys}\Delta T = q_{sys} = -q_{surr} = -mC_{H2O}\Delta T - C_{cal}\Delta T$$

This means we also need to know the heat capacity of the calorimeter itself. Usually, when you purchase one, the C_{cal} is provided to you, and that is what we do in calculations in this course. We also assume that the change in temperature in the water also applies to the change in temperature of the calorimeter.

Question 2 1 pts

Which of the following are essential components of a bomb calorimeter?

A mechanical stirrer

A heavy, fine threaded system container

A thermometer

All of these answers are correct.

Question 3 1 pts

Which is the more likely statement:

The calculation of energy in the surroundings is less than the actual system energy.

or

The calculation of energy in the surroundings is more than the actual system energy.

less

more

Question 4 1 pts

Suppose ethanol is used instead of water as the solvent in the container surrounding the system. If ethanol has a heat capacity that is about half that of water, what will change in the calorimetry calculation?

There will be no change in the calculation.

The change in temperature of the surroundings will be half that if water is used.

The change in temperature of the surroundings will double.

Question 5 1 pts

Ten potato chips are combusted in a bomb calorimeter. 1000 grams of water increases in temperature from 25^{o}C to 45 ^{o}C. What is the total heat content of the ten potato chips?

- 100 kJ
- 200 kJ
- 40 kJ
- 80 kJ

Question 6 1 pts

Referring to the problem above, how many kJ are in a potato chip?

- 10 kJ
- 20 kJ
- 8 kJ
- 4 kJ

Coffee Cup Calorimetry

Coffee cup calorimeters are just what they sound like, coffee cups. They are open systems where the top is open to the environment. The consequences of this is that gases are allowed to expand out of the system, so the work done by the system is NOT equal to zero. Since the pressure is constant from the atmosphere:

$$\Delta H = q_p$$

Most of you have probably done a coffee cup calorimetry measurement at some point in your life. All you need is a cup, a thermometer and some water. Usually, the water serves not just as the place where the heat goes to change the temperature, but also as the system itself. For example, you might decide to measure the amount of heat associated with adding salt to water. If the process is exothermic, the temperature will go up a few degrees and if you know the mass of the water, you can quickly calculate $\Delta H = q_p$.

What is most important to appreciate about the coffee cup calorimeter is not that it is an inexpensive way to learn about calorimetry in a science lab, but rather that it represents the actual amount of heat generated in an open system where the pressure is constant. In the next module, we will learn to work lots of thermodynamics calculations, and they will all be calculations of ΔH, not ΔE.

Question 7 1 pts

Some sodium chloride is added to 100 grams of water in a styrofoam cup. A thermometer is used to stir the salt water while the temperature is measured. The temperature decreases from 25^{o}C to 24 ^{o}C. What can be said about the thermodynamics of dissolving NaCl in water?

- NaCl dissolving in water is exothermic.
- Something is wrong with the experiment. Calorimetry experiments have to have an increase in temperature.
- NaCl dissolving in water is an endothermic process.

Calorimetry in the real world.

You may have noticed that we talked about calorimetry, but never talked about calories. This is because scientists like to work in SI units, and the SI units of energy are Joules.

But there is a simple conversion from Joules to calories:

4.184 J = 1 calorie.

The other interesting thing to know is that a single Joule or calorie is not a lot of energy. We need to put it on a scale consistent with the amount of food we eat.

So when you see that a snack has 100 calories on the label, it actually has 100 kilocalories. Maybe this lie on a package is done to save space, or maybe to avoid folks having to learn the metric system, or maybe no one wants to eat a bag of chips thinking that they just ate 100,000 calories.

When you perform a bomb calorimetry measurement on a banana and find out that 400,000 J of energy are released, you need to first convert it into kJ and then into kcal, and only then will you have the 100 calories in a banana you know in the real world.

[Video Transcript (https://onramps.instructure.com/courses/4073013/files/292195383?wrap=1)](https://onramps.instructure.com/courses/4073013/files/292195383?wrap=1) (https://onramps.instructure.com/courses/4073013/files/292195383/download?download_frd=1)

Gummy bears pack a lot of energy.

In this video, Dr. Laude burns a gummy bear in potassium chlorate in order to demonstrate how much energy is contained in a gummy bear. What you should take away is how much energy is contained in everyday food and how much work we would need to do in order to burn off the calories we consume. In this case, quite a lot of energy is contained in just 1 gummy bear.

Question 8 1 pts

A bomb calorimetry measurement indicates that a single potato chip has a heat content of 20,000 J. What is on the label of a bag of potato chips that contains 20 chips?

- 5000 calories
- 5 calories
- 100,000 calories
- 100 calories

Calculating ΔH Across Phase Changes

As we finish up our discussion of the first law it is instructive to ask about something we see in everyday life that ties together what we have learned.

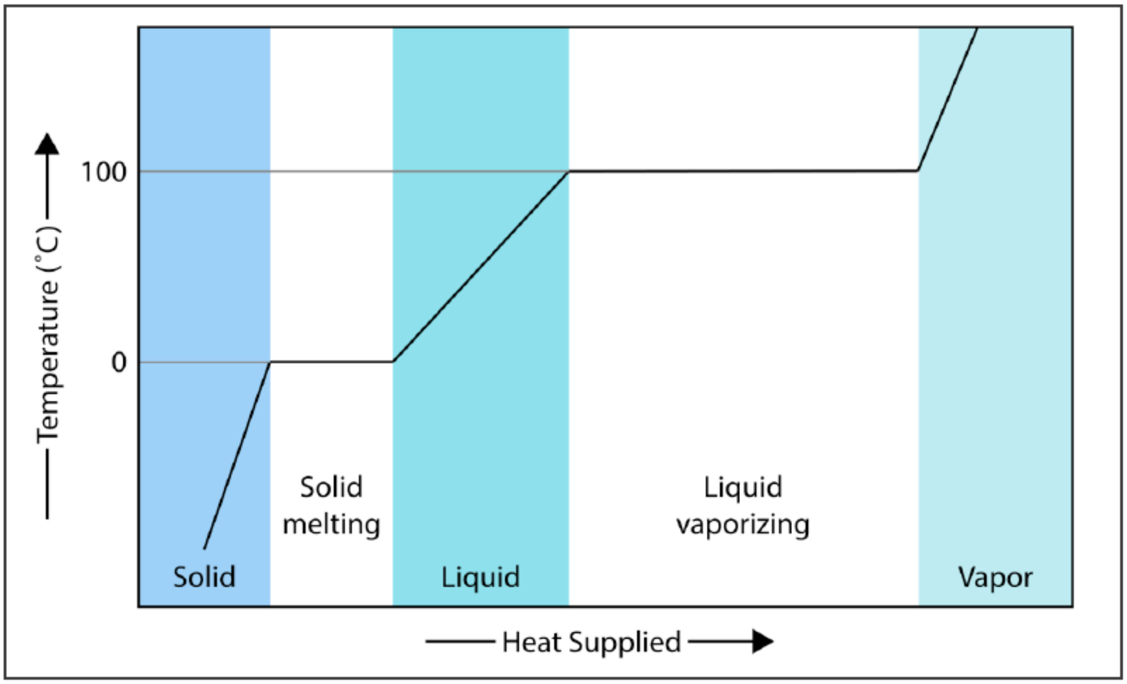
How much energy is necessary to convert a block of frozen ice into water vapor? This is an easy question to visualize:

Take a piece of ice out of the freezer, put it in a pot on the stove, and heat it up until it has all evaporated.

If you think about it, there are five distinctly different things going on during the process of turning ice into steam:

- ice warming up to 0^oC after you take it out of the freezer
- ice melting at 0^oC to make liquid water
- water warming from 0^oC to 100^oC
- water vaporizing at 100^oC
- water vapor heating above 100^oC.

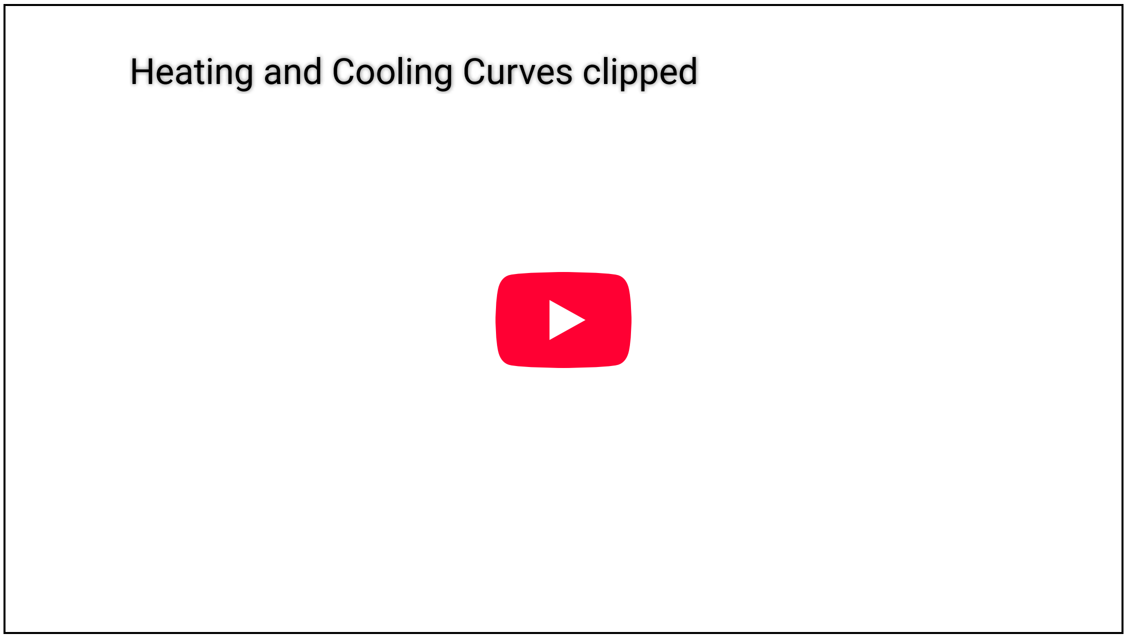
Put them altogether and you have the heating curve diagram below in which the amount of heat added is on the x axis as the temperature rises.



Don't continue reading until you have visualized each of these events!!

In the video above, Dr. Laude explained the important features of a heating curve and its five regions. It contains a couple of flat lines where equilibrium exists: two flat lines represent phase transitions where the temperature doesn't change and three warming regions where the temperature rises in each of the three phases.

The math behind heating curves is actually really simple, but can seem complicated because there are 10 variables to take into account.   Be careful to keep track of what is going on in the system and what variables are needed to complete the calculations.



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Sample problem: How much heat does it take to raise the temperature of 10 grams of ice at -20°C to water vapor at 130°C?

- Here are the constants you will need (the constants are all rounded off to make the math easier):
- C_{ice}: 2 J/g°C
- C_{water}: 4 J/g°C
- C_{steam}: 2 J/g°C
- ΔH_{fus}: 300 J/g
- ΔH_{vap}: 2000 J/g

Now it is time to solve the problem, one region at a time.

Region 1: 10 grams of ice warming

$\Delta H_{ice\ warming} = mC\Delta T$ = 10 grams x 2 J/g°C x 20°C = **400 J**

Region 2: 10 grams of ice melting

$\Delta H_{ice\ melting}$ = 10 grams x 300 J/g = **3000 J**

Region 3: 10 grams of water warming

$\Delta H_{water\ warming} = mC\Delta T$ = 10 grams x 4 J/g°C x 100°C = **4000 J**

Region 4: 10 grams of water vaporizing

$\Delta H_{\text{water vaporizing}}$ = 10 grams x 2000 J/g = **20000 J**

Region 5: 10 grams of steam warming

$\Delta H_{\text{steam warming}}$= $mC\Delta T$ = 10 grams x 2 J/g^{o}C x 30^{o}C = **600 J**

Adding them all together:

ΔH_{total} = 300J + 3000J   + 4000J + 20,000J + 600J = 27900 or 27.9 kJ.

Things to note:

- Don't worry about how long it took to work this problem. On the exam you might be asked to calculate for just one or two regions
- You worked a calorimetry problem three times ($mC\Delta T$) just like in the previous module
- Rationalizing why so much energy went into vaporizing the water--the answer is that this is when the strong H bonding IMF are broken.

Quiz saved at 7:48pm Submit Quiz